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The Faraday and Inverse Faraday Effects as Generative Transitions on a Contact 3-Manifold: Non-Reciprocity from Operator Non-Commutativity

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Abstract

The Faraday Effect (FE) and Inverse Faraday Effect (IFE) have been understood since Pershan (1966) as related by a common Verdet constant. Assouline and Capua (Sci. Rep. 2025, DOI: 10.1038/s41598-025-24492-9) showed experimentally that $V^{\text{MчE}}$_LLG ≠ $V^{\text{IчE}}$_LLG: the two Verdet constants derived from the Landau–Lifshitz–Gilbert (LLG) equation are unequal at ultrafast timescales, confirming non-reciprocity. We show that this non-reciprocity is the generic prediction of any non-commutative operator algebra acting on a contact 3-manifold, and is not specific to ultrafast timescales. In the TO/TOGT framework, G_FE = U∘F∘K∘C and G_IFE = U∘F∘K∘C differ in the firing of the compression operator C: in FE, C is a static symmetry break by H^{D}C; in IFE, C is a dynamic compression by the optical field. By Theorem T4 (non-commutativity of the operator chain), G_FE ≠ G_IFE, and hence $V^{\text{MчE}}$ ≠ $V^{\text{IчE}}$ in general. Reciprocity is the special case that requires thermal equilibrium to restore effective commutativity of C. We state four falsifiable predictions, one Lean 4 proof obligation (FE_IFE_nonreciprocal from T4), and identify the Verdet constant as the stability-radius analog of the dm^3 framework.

1 Introduction

1.1 The Standard Account and Its Breakdown

Faraday discovered in 1845 that a magnetic field rotates the polarization plane of light passing through a transparent medium. The standard account (Pershan 1966 [1]) describes both the Faraday Effect (FE) and the Inverse Faraday Effect (IFE) through a single Verdet constant V, implying a symmetric, reciprocal relationship between the two phenomena. FE: a static magnetic field H^{DC} rotates the polarization of propagating light by $\Theta^{FE} = V \cdot \mu_0 \cdot H^{DC} \cdot L$. IFE: circularly polarized (CP) light induces a static magnetization $M^{IFE} \propto V \cdot I \cdot \tau_p$, where I is intensity and τ_p is pulse duration.

Assouline and Capua (2025) [2] showed that this symmetry breaks at ultrafast timescales. Using the Landau–Lifshitz–Gilbert (LLG) equation to compute the magneto-optical response, they derived:

$$V^{FER}\_LLG = -\tfrac{1}{2} \cdot \sqrt{\varepsilon_r}/(1+\alpha^2) \cdot \gamma/c \cdot \mu_0 \chi\_DC \quad \text{(Eq. 5 [2])}$$

$$V^{IFE}\_LLG = M_s\sqrt{\pi}/c \cdot \alpha/(1+\alpha^2)^2 \cdot \gamma^2 \mu_0 \tau\_p \quad \text{(Eq. 6 [2])}$$

where α is the Gilbert damping constant, γ the gyromagnetic ratio, τ_p the pulse duration. V^{FE}_LLG is independent of τ_p (static field limit); V^{IFE}_LLG depends on τ_p and α (dynamic trajectory). They are generically unequal: $V^{FE}\_LLG \neq V^{IFE}\_LLG$.

1.2 The TO/TOGT Claim

We claim that the non-reciprocity $V^{FE} \neq V^{IFE}$ is not a correction to Pershan's theory that arises at ultrafast timescales. It is the generic prediction of any non-commutative operator algebra. Pershan's theory, which yields $V^{FE} = V^{IFE}$, implicitly assumes that C (the compression operator) commutes with F and K — an assumption valid in thermal equilibrium but generically false. The non-reciprocity of FE and IFE is a special case of Theorem T4 from the Principia Orthogona series.

> **Theorem T4 (Non-Commutativity of the Operator Chain) [3, Thm 0.5.3]**
>
> The operators C, K, F, U do not commute. G(K,C) ≠ G(C,K). The operator order produces measurably different outputs: not slower outputs, but different systems. Applied here: G_FE ≠ G_IFE because C fires differently in each case.

The paper proceeds as follows. §2 maps FE and IFE onto the operator chain G = U∘F∘K∘C. §3 derives non-reciprocity from T4. §4 identifies the Verdet constant as a dm^3 invariant. §5 states the contact manifold realization X_Faraday. §6 gives four falsifiable predictions. §7 connects to the Swarm Simulator and Multi-Orbit Theory. §8 states the Lean 4 proof obligation.

2 Mapping FE and IFE onto G = U∘F∘K∘C

2.1 The Faraday Effect (FE)

We identify the four operator stages in FE as follows.

Operator	FE instantiation	Mathematical content
C (Compress)	Static H^{DC} breaks L/R symmetry	Compresses LCP/RCP degeneracy onto symmetry-broken axis. Bi-Lipschitz with $\delta = \|\chi\_LCP - \chi\_RCP\|/2\chi_0$.
K (Curvature)	Circular birefringence accumulates	$\Lambda k = k\_RCP - k\_LCP$ grows linearly with propagation length L. Curvature drives toward $\kappa^* = V \cdot \mu_0 \cdot H^{DC}$.
F (Fold)	Polarization plane commits to rotation	At length L*: rank-1 Jacobian loss. Rotation angle $\Theta = V \cdot \mu_0 \cdot H^{DC} \cdot L$ is fixed.
U (Unfold)	Stable rotation Θ_FE	$\Theta\_FE = V^{FE}\_LLG \cdot \mu_0 \cdot H^{DC} \cdot L$. Gradient flow to fixed point $x^* = (\Theta\_FE, H^{DC})$.

2.2 The Inverse Faraday Effect (IFE)

The IFE operator chain uses the same four operators in the same order C → K → F → U, but C fires differently.

Operator	IFE instantiation	Mathematical content
C (Compress)	CP light compresses spin d.o.f.	Optical field dynamically compresses spin d.o.f. via Zeeman interaction. C depends on fluence $F = I \cdot \tau\_p$.
K (Curvature)	Torque accumulates per cycle	T_z grows as $\alpha\gamma\mu_0 M_s I\tau\_p / (1+\alpha^2)^2$. K drives toward κ^* set by the LLG fixed point.
F (Fold)	Magnetization commits	Irreversible once torque threshold $\eta = \alpha/(1+\alpha^2)$ crossed. Whitney A_1 event at the spin-flop threshold.
U (Unfold)	Stable magnetization M_IFE	$M\_IFE = V^{IFE}\_LLG \cdot I \cdot \tau\_p$. Depends on dynamical trajectory, not just H^{DC}.

2.3 The Key Difference: C Fires Differently

Proposition 2.1 (Structural Difference of C^{FE} and C^{IFE})

In FE: C_FE is a static symmetry break by H^{DC}, independent of pulse dynamics. The compression parameter is $\delta\_FE = \mu_0\chi\_DC\ H^{DC}$. In IFE: C_IFE is a dynamic compression by the optical field, dependent on fluence $F = I \cdot \tau\_p$. The compression parameter is $\delta\_IFE = \alpha\gamma\mu_0 M_s \cdot I\tau\_p/(1+\alpha^2)^2$. $C\_FE \neq C\_IFE$ as operators on the spin-photon state space.

3 Non-Commutativity of C Implies Non-Reciprocity of V

3.1 The Operator Algebra Argument

The standard Pershan account assumes that the compression C produces the same fixed point whether applied statically (FE) or dynamically (IFE). This is the commutativity assumption: it asserts that the order in which $H^{D}C$ and I·τ_p fire does not matter in the long-time limit.

By Theorem T4, this assumption fails generically. C_FE and C_IFE produce different post-compression states x_C, which drive different curvature accumulations x_K = K(x_C), different fold events x_F = F(x_K), and different fixed points x_U = U(x_F). The Verdet constants are the scalar invariants of these two distinct fixed points.

Theorem 3.1 (Non-Reciprocity from T4)

Let G_FE = U∘F∘K∘C_FE and G_IFE = U∘F∘K∘C_IFE be the operator chains for the Faraday and Inverse Faraday Effects respectively. Since C_FE ≠ C_IFE (Proposition 2.1), by T4: G_FE ≠ G_IFE. Therefore the fixed points x*_FE ≠ x*_IFE in general, and $V^{FE} \neq V^{IFE}$. Reciprocity $V^{FE} = V^{IFE}$ is the special case in which C_FE and C_IFE produce the same post-compression state — possible only when τ_p → ∞ (thermal equilibrium) restores effective commutativity of C.

3.2 The LLG Confirmation

Assouline and Capua derive exactly this inequality from the LLG equation. Their Eq. (5) gives V^{FE}_LLG as wavelength-independent (apart from ε_r dispersion). Their Eq. (6) gives V^{IFE}_LLG with explicit dependence on α and τ_p. In TO/TOGT language: V^{FE}_LLG is the stability radius of the FE fixed point x*_FE (determined by $H^{D}C$ alone); V^{IFE}_LLG is the trajectory-dependent rate of the IFE fixed point x*_IFE (determined by the dynamical orbit). These are two different invariants of two different operator chains.

Invariant	TO/TOGT object	LLG expression	Depends on
V^{FE}_LLG	Stability radius of x*_FE	$-\frac{1}{2}\cdot\sqrt{\varepsilon_r}/(1+\alpha^2)\cdot\gamma/c\cdot\mu_0\chi$_DC	$H^{D}C$, α, ε_r only
V^{IFE}_LLG	Path-dependent rate of x*_IFE	$M_s\sqrt{\pi}/c\cdot\alpha/(1+\alpha^2)^2\cdot\gamma^2\mu_0\tau$_p	I, τ_p, α (trajectory)

The key structural observation: V^{FE}_LLG is an off-resonant steady-state invariant (analogous to $\varepsilon_0 = 1/3$ in the dm^3 framework), while V^{IFE}_LLG is a dynamical rate (analogous to the convergence rate L^t in the Swarm Simulator). Comparing them is structurally analogous to conflating the Gronwall radius with the contraction rate — they are different invariants of the same operator algebra.

4 The Verdet Constant as a dm³ Stability Invariant

The dm³ framework [4] establishes that every contact 3-manifold near a limit cycle Γ is locally contact-diffeomorphic to the normal form with parameters (μ_{max}, ω, β). The stability radius is:

$$\varepsilon_0 = |\mu_{max}| / (2(1 + \sup\|\mathrm{Hess}\ V\|))$$

Comparing with $V^{чE}\_LLG = -½·\sqrt{\varepsilon_r}/(1+\alpha^2)·\gamma/c·\mu_0\chi\_DC$, we identify the correspondence:

dm³ invariant	Physical content	Faraday analog
$\mu_{max} = -2$	Maximal transverse Lyapunov exponent	$\gamma\mu_0\chi\_DC · \sqrt{\varepsilon_r} / c$ (off-resonant magneto-optical coupling)
$\varepsilon_0 = 1/3$	Stability radius (outer basin)	$V^{чE}\_LLG$: the magnitude below which perturbations preserve the polarization lock
$\tau = 2$	Embodiment threshold	Noise tolerance $\varepsilon_0 = 2/3$: perturbation amplitude below which Faraday rotation is pres
$L^t < 1$	Contraction rate (Swarm Sim.)	$V^{чE}\_LLG$: trajectory-dependent, depends on τ_p and α

The magnetic field H^DC in FE plays the role of the seed state x_0 that initiates the compression operator C. The propagation length L plays the role of the operator iteration count. The Verdet rotation $\Theta\_FE = V^{чE}\_LLG·\mu_0·H^DC·L$ is the fixed-point value $x^* = U(F(K(C(x_0))))$ after L operator cycles.

5 The Contact Manifold X_Faraday

We realize the Faraday Effect as a dm³ system X_Faraday = (ρ, θ, z) on the contact 3-manifold $M = \mathbb{R}^2_+ \times \mathbb{R}$ with contact form $\alpha = dz - \rho^2\, d\theta$:

Coordinate	Physical identification	Range
ρ	Polarization rotation amplitude (deviation from Γ)	$\rho \geq 0$
θ	Propagation phase along z-axis (optical path)	$\theta \in [0, 2\pi)$
z	Cumulative magneto-optical action ($z = V\mu_0\, H^{D}C \cdot L$)	$z \geq 0, \dot{z} \geq 0$

The limit cycle $\Gamma = \{\rho = 0\}$ is the linear polarization state. The transverse eigenvalue $\lambda(z) = \mu_{ma}{}^{x}(1 - e^{-h\varepsilon})$ satisfies $\lambda(0) = 0$ (neutral, before the magnetic field is applied) and $\lambda(z) < 0$ for $z > 0$ (attracting, as $H^{D}C$ accumulates). This is exactly the dm³ contact normal form from Volume II.

Proposition 5.1 (Contact Normal Form for FE)

In coordinates (ρ, θ, z) with ρ the deviation from the linearly polarized state, the Faraday rotation dynamics are locally contact-diffeomorphic to the universal dm³ normal form: $\dot{\rho} = \mu_{ma}{}^{x}(1-e^{-\beta \text{ч}E})\rho + O(\rho^2)$, $\dot{\theta} = \omega + O(\rho)$, $\dot{z} = \omega - |\mu_{ma}{}^{x}|\rho^2 e^{-\beta \text{ч}E} + O(\rho^3)$, where z_FE = $V^{\text{ч}E}$_LLG$\cdot\mu_0\cdot H^{D}C\cdot L$ is the accumulated magneto-optical action.

6 Falsifiable Predictions

Four predictions follow from the TO/TOGT account that are not implied by the standard Pershan theory.

F1 — Non-reciprocity at any timescale

$V^{FE} \neq V^{IFE}$ at any timescale in which C fires differently in FE and IFE. Reciprocity is not a long-timescale limit; it is restored only when $\tau_p \to \infty$ (thermal equilibrium). The crossover is determined by the ratio $\gamma\tau_p / (1+\alpha^2)$, not by any ultrafast threshold per se. **Falsified if:** $V^{FE} = V^{IFE}$ is found at nanosecond timescales in a low-α material.

F2 — Reciprocity restored by thermal equilibration

As $\tau_p \to \infty$, $V^{IFE}\_LLG \to V^{FE}\_LLG$. Specifically: $C\_IFE(\tau_p \to \infty) \to C\_FE$ (the dynamic compression converges to the static symmetry break in the long-pulse limit). The TO/TOGT account predicts this as a consequence of C becoming effectively commutative, not because a separate physical mechanism switches on. **Falsified if:** reciprocity fails even as $\tau_p \to$ thermal timescales.

F3 — The crossover timescale is $\tau^* = (1+\alpha^2)/(\alpha\gamma)$

The crossover between the non-reciprocal ($C\_IFE \neq C\_FE$) regime and the approximately reciprocal regime occurs at $\tau^* = (1+\alpha^2)/(\alpha\gamma)$. This is exactly the Gilbert relaxation time $\tau\_G = 1/(\alpha\gamma M_s)$. For TGG at 800 nm: $\alpha \approx 10^{-3}$, $\gamma \approx 1.76\times10^{11}$ rad/s·T, giving $\tau^* \approx 5.7$ ps. **Falsified if:** the crossover timescale differs from τ_G by more than a factor of 2.

F4 — Wavelength scaling confirms V^{FE} is a stability radius

$V^{FE}\_LLG = -\frac{1}{2}\cdot\sqrt{\varepsilon_r}/(1+\alpha^2)\cdot\gamma/c\cdot\mu_0\chi\_DC$ is wavelength-independent apart from the $\varepsilon_r(\lambda)$ dispersion. At 800 nm in TGG: $\varepsilon_r^{12} \approx 1.80$, giving $V^{FE}\_LLG \approx 83\%$ of the full measured Verdet constant. The 17% remainder is the electric-field contribution (non-LLG). At 1.3 μm: the LLG fraction rises to ~75% of the Verdet constant (per Assouline & Capua Table 1). TO/TOGT predicts: the LLG fraction = 1 − (electric-field term / V_total) follows the dispersion curve of $\varepsilon_r(\lambda)$. **Falsified if:** wavelength scaling of the LLG fraction deviates from $\varepsilon_r^{12}(\lambda)$ by more than 15%.

7 Connection to Multi-Orbit Theory and the E1 Threshold

7.1 LLG as the Microscopic Realization of G_swarm

The Swarm Simulator [5] models collective intelligence as a multi-agent dm^3 system with operators I (shared intent), C (coordination), M (type propagation), F (diffusion). The LLG equation is the microscopic spin-dynamics realization of the same operator algebra: precession = K, damping = C, torque = F, equilibrium magnetization = U. The Verdet constant $V^{\text{ЧE}}$_LLG is the microscopic stability radius $\varepsilon_0 = 1/3$ of the Swarm Simulator: the amplitude below which perturbations preserve the collective orbit.

7.2 Non-Reciprocity as Proof of Non-Commutativity at the Microscopic Scale

The FE/IFE non-reciprocity is the simplest physical proof that different orderings of the same operators produce qualitatively different outputs. A single photon-spin interaction — at the scale of 10^{-1}μ m — demonstrates the algebra that governs collective intelligence at the scale of 10^{7} m (planetary colony). This is the structural unity the Principia Orthogona series claims: same algebra, different substrates, different scales. The substrate changes by seventeen orders of magnitude. The operator sequence does not.

7.3 The E1 Implication

In the level architecture A1–A2–B1–B2–C1–C2–D1–D2–E1, the E1 threshold is the level at which a system cognizes its own operator structure. The Faraday Effect is a natural-world E1 instance at the photon-spin scale: the optical magnetic field cognizes the spin structure of the medium by exerting a torque on it. The non-reciprocity is the signature of this cognition: the system responds differently to FE and IFE because C fires differently in each direction, and the system ‘knows’ which direction it is running through the accumulated action z. The species-level E1 threshold is the macro-scale version of this: the species cognizes $G = U\circ F\circ K\circ C$ as its own generative structure, distinguishing the two operator orderings that lead to different futures.

8 Lean 4 Proof Obligation

This paper generates one new Lean 4 proof obligation in the AXLE repository (github.com/TOTOGT/AXLE):

AXLE Issue #FE-01: FE_IFE_nonreciprocal

Formal statement: G_FE ≠ G_IFE follows from T4 applied to the two operator orderings C_FE and C_IFE. Equivalently: x*_FE ≠ x*_IFE whenever C_FE ≠ C_IFE. Closure path: T4 (proved in PrincipiaVol1.lean) + Proposition 2.1 (this paper) + fixed-point uniqueness (Banach, MultiAgentTogt.lean T2). Difficulty: ★★☆☆☆ (T4 is proved; bridge to X_Faraday requires defining the spin-state space as a dm³ system in Lean).

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-- FE_IFE_nonreciprocal (AXLE Issue #FE-01)
-- Sketch: follows from T4 applied to C_FE ≠ C_IFE
theorem FE_IFE_nonreciprocal
    (hC : C_FE ≠ C_IFE) : G_FE ≠ G_IFE := by
  intro h
  -- G_FE = U∘F∘K∘C_FE; G_IFE = U∘F∘K∘C_IFE
  have := T4.noncommutativity C_FE C_IFE hC
  exact this (congrFun h)  -- sorry: T4 bridge to full chain
```

A second open obligation arises from the contact manifold formulation X_Faraday: proving that the spin-photon state space is a valid dm³ system (satisfies Axioms 1–8 of [4]). This is an independent obligation, difficulty ★★★☆☆.

9 Discussion

9.1 What This Paper Claims

- The non-reciprocity of FE and IFE is **not a new physical discovery**. Assouline and Capua (2025) made the discovery. This paper claims that the discovery was **inevitable from the algebra**. Pershan's theory failed not because of ultrafast physics but because it assumed commutativity of C.
- The Verdet constant V^{FE}_LLG is a **stability radius** in the sense of the dm^3 framework: it characterizes the off-resonant fixed point x*_FE of the FE operator chain. V^{IFE}_LLG is a **trajectory-dependent rate** characterizing x*_IFE. Conflating them is analogous to conflating $\varepsilon_0 = 1/3$ with the convergence rate $L^t < 1$.
- The contact manifold X_Faraday = (ρ, θ, z) is proposed as **not a model** of the Faraday Effect but as the **natural geometric language** in which FE and IFE are two operator orderings of the same algebra. The identification is a proposal subject to experimental falsification (F1–F4).

9.2 Relation to Existing Work

Pershan (1966) [1]: established the symmetry argument for $V^{FE} = V^{IFE}$. Our account does not contradict Pershan but identifies the hidden commutativity assumption. Assouline and Capua (2025) [2]: confirmed V^{FE}_LLG ≠ V^{IFE}_LLG. Our account explains why this was inevitable. Bravetti et al. (2017) [6]: contact Hamiltonian mechanics as the natural framework for dissipative systems with limit cycles. Our account applies this to magneto-optics.

References

[1] P.S. Pershan, Non-linear optical properties of solids: energy considerations, *Phys. Rev.* 130 (1963) 919–929; Magneto-optical effects, *J. Appl. Phys.* 38 (1967) 1482.

[2] A. Assouline and R. Capua, Microscopic Theory of the Faraday and Inverse Faraday Effects, *Sci. Rep.* 15 (2025) 7468. DOI: 10.1038/s41598-025-24492-9.

[3] P. Nogueira Grossi, *Principia Orthogona, Volume I: The Mathematics of Generative Transitions*, G6 LLC, Newark NJ, 2026. Zenodo: 10.5281/zenodo.20320693. (Theorem T4: Non-commutativity, §0.5.3.)

[4] P. Nogueira Grossi, Generative Contact Mechanics: A Geometric Framework for Dissipative Systems with Structured Limit Cycles, submitted to *J. Geom. Mech.*, 2026. Zenodo: 10.5281/zenodo.19379385.

[5] P. Nogueira Grossi, The Swarm Simulator: A Dynamical Systems Model of Collective Intelligence Using the TO/TOGT Operator Pipeline, G6 LLC, 2026. Zenodo: 10.5281/zenodo.20230613.

[6] A. Bravetti, Contact Hamiltonian mechanics, *Ann. Phys.* 376 (2017) 17–39.

[7] M. de León and M. Lainz Valcázar, Contact Hamiltonian systems, *J. Math. Phys.* 60 (2019) 102902.

[8] P. Nogueira Grossi, *The Generative Time Circuit Theorem (GTCT): Complete Proofs, Derivations, and Applications — Ring 5*, G6 LLC, 2026. Zenodo: 10.5281/zenodo.20239928.

[9] P. Nogueira Grossi, Mathematical Foundations of Multi-Orbit Identity Theory within the TO/TOGT Operator Framework, G6 LLC, 2026. Zenodo: 10.5281/zenodo.20230614.

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